

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program/Batch	B082-Networkwalks
Date	21 August 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools and scanning my own local network with Zenmap. One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the technical results observed, and an analysis of why each finding matters from a security assessment perspective.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance and scanning activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Windows CMD	Local IP and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon. Each tool was used to collect a different type of information about the target.

- WHOIS Enumeration:** Used WHOIS to obtain domain registration information and authoritative nameservers. The scan identified registrar GoDaddy.com, LLC (IANA ID 146), creation date 2019-11-06, expiration date 2027-11-06, privacy masking via Domains By Proxy LLC, and delegation to HostGator nameservers (NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM). DNSSEC is confirmed as unsigned.

```
(kali@kali)-[~]
$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-08-21T04:39:19Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
Domain Name: networkwalks.com
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: +14806242505
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Registry Registrant ID: Not Available From Registry
Registrant Name: Registration Private
Registrant Organization: Domains By Proxy, LLC
Registrant Street: 180 S. Mill Ave, Suite 1600
Registrant City: Tempe
Registrant State/Province: Arizona
Registrant Postal Code: 85281
Registrant Country: US
Registrant Phone: +14806242599
Registrant Phone Ext:
Registrant Fax:
Registrant Fax Ext:
Registrant Email: https://www.godaddy.com/whois/results.aspx?domain=networkwalks.com&action=contactDomainOwner
Tech Name: Registration Private
Tech Organization: Domains By Proxy, LLC
Tech Street: DomainsByProxy.com
Tech City: Tempe
Tech State/Province: Arizona
Tech Postal Code: 85281
Tech Country: US
Tech Phone: +14806242599
Tech Phone Ext:
Tech Fax:
Tech Fax Ext:
Tech Email: https://www.godaddy.com/whois/results.aspx?domain=networkwalks.com&action=contactDomainOwner
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
Name Server: NS38.DOMAINCONTROL.COM
Name Server: NS38.DOMAINCONTROL.COM
DNSSEC: unsigned
URL of the ICANN Whois Data Problem Reporting System: http://wdprs.internic.net/
```

- **WhatWeb Fingerprinting:** I used WhatWeb to fingerprint technologies on networkwalks.com. The output revealed that the site is running on an Apache HTTP server located at IP 192.232.216.135 (US). It identified WordPress 7.1, Bootstrap 7.1, jQuery 3.7.1, WordPress Download Manager plugin v3.3.58, administrative contact info (info@networkwalks.com), and custom caching headers (x-nginx-cache: WordPress, x-endurance-cache-level: 0).

```
(kali@kali)-[~]
$ whatweb networkwalks.com
https://networkwalks.com [301 Moved Permanently] Apache, Cookies[_wpdm_client], Country[UNITED STATES][us], HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1], Cookies[_wpdm_client], Country[UNITED STATES][us], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&prtVar2=76&prtVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1], Country[UNITED STATES][us], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&prtVar2=76&prtVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
```

- **Nslookup DNS Resolution:** Using Nslookup with resolver 8.8.8.8, I resolved networkwalks.com to IPv4 address 192.232.216.135, confirming the hosting address of the primary web application.

```
(kali@kali)-[~]
$ nslookup networkwalks.com
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
```

- **cURL Header Analysis:** I used Curl with the -I option to inspect HTTP response headers. The output confirmed an HTTP/2 200 OK status, Apache server banner, WordPress caching headers, secure cookie attributes (__wpdm_client), and explicit Link headers revealing the WordPress REST API endpoint (/wp-json/ and /wp-json/wp/v2/pages/53).

```
(kali@kali)-[~]
$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=0d2c01b3bddc920eee8357df61a734ed; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Fri, 21 Aug 2026 04:56:16 GMT
server: Apache
```

- **Wafw00f WAF Detection:** I executed Wafw00f to identify perimeter Web Application Firewalls. The scan detected ModSecurity (SpiderLabs) protecting the target web application.

```
(kali@kali)-[~]
$ wafw00f networkwalks.com

      ( Woof! )
    /  \  /  \
   /    \    \
  /      \    \
 /        \    \
/          \    \
~ WAFW00F : v2.4.2 ~
The Web Application Firewall Fingerprinting Toolkit

[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2

(kali@kali)-[~]
$
```

- **DNSRecon Zone Enumeration:** I used DNSRecon to perform comprehensive zone and service record enumeration. The scan identified HostGator authoritative nameservers (50.87.144.87, 192.232.216.131), MX records pointing to mail.networkwalks.com (192.232.216.135), SPF TXT records (v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all), 8 cPanel autodiscover SRV records, and exposed the BIND daemon version string ('9.16.23-RH').

```
(kali@kali)-[~]
$ dnsrecon -d networkwalks.com
2026-08-21T01:01:20.238528-0400 INFO Starting enumeration for domain: networkwalks.com
2026-08-21T01:01:20.239764-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-08-21T01:01:21.059430-0400 ERROR No answer for DNSSEC query for networkwalks.com
2026-08-21T01:01:21.826168-0400 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-08-21T01:01:22.965867-0400 INFO NS ns6135.hostgator.com 50.87.144.87
2026-08-21T01:01:23.538771-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-08-21T01:01:23.539809-0400 INFO NS ns6136.hostgator.com 192.232.216.131
2026-08-21T01:01:24.100061-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-08-21T01:01:24.907029-0400 INFO MX mail.networkwalks.com 192.232.216.135
2026-08-21T01:01:25.345807-0400 INFO A networkwalks.com 192.232.216.135
2026-08-21T01:01:26.298258-0400 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnzDNVK4q75X-Ij-mjgEeg-UsYI
2026-08-21T01:01:26.299027-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-08-21T01:01:27.130282-0400 INFO Enumerating SRV Records
2026-08-21T01:01:31.129618-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.11 443
2026-08-21T01:01:31.130062-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.14 443
2026-08-21T01:01:31.130481-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.15 443
2026-08-21T01:01:31.130755-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.9 443
2026-08-21T01:01:31.130976-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.15 443
2026-08-21T01:01:31.131500-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.11 443
2026-08-21T01:01:31.132042-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.9 443
2026-08-21T01:01:31.132490-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.14 443
2026-08-21T01:01:31.134522-0400 INFO 8 Records Found
2026-08-21T01:01:31.135568-0400 INFO Completed enumeration for domain: networkwalks.com

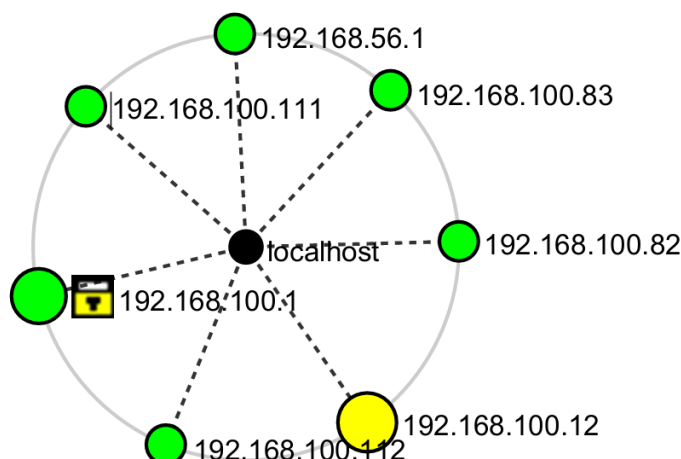
(kali@kali)-[~]
$
```

4.2 Network Scanning with Zenmap

For the second activity (Module W2-PM5), I used Zenmap (official GUI front-end for Nmap) on my Windows PC to perform host discovery and topology mapping on my local LAN:

- **Task 1 (Installation):** Downloaded and installed Zenmap with Npcap on Windows.
- **Task 2 (Subnet Identification):** Executed ipconfig in Windows Command Prompt, identifying the local IP address 192.168.56.1 and LAN subnet mask 255.255.255.0.
- **Task 3 (Live Host Discovery):** Configured Zenmap with target 192.168.56.1/24 and executed a Ping Scan (nmap -sn 192.168.56.1/24).
- **Task 4 (Host Count):** A total of six (6) hosts were identified as live on the local subnet (including the scanning PC).

- **Task 5 (Live IP Addresses):** The live IP addresses discovered were: 192.168.56.1, 192.168.100.82, 192.168.100.83, 192.168.100.111, 192.168.100.112, and 192.168.100.12.
- **Task 6 (MAC Addresses & Vendors):** Submitted on networkwalks.com
- **Task 7 (Topology Generation):** Navigated to the Zenmap Topology tab, turned on the legend, and saved the network topology graph in PDF format.



5. Risk Analysis / Impact

Based on the technical data gathered from the footprinting and network scanning activities, I analyzed the following security observations and potential attack surface risks:

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	DNS Daemon Software Version Disclosure	DNSRecon disclosed BIND Version '9.16.23-RH' on authoritative nameservers	Allows adversaries to query vulnerability databases for version-specific CVEs affecting BIND 9.16.23-RH, facilitating targeted service disruption or exploit planning.	● Medium (CVSS: 5.3)
2	CMS & Plugin Version Information Exposed	WhatWeb identified WordPress 7.1 and WP Download Manager 3.3.58	Exposing specific CMS and plugin versions allows attackers to identify unpatched vulnerabilities or targeted proof-of-concept exploits without blind fuzzing.	● Medium (CVSS: 4.3)
3	WordPress REST API Endpoint Exposure	cURL response headers exposed /wp-json/ and /wp-json/wp/v2/pages/53	Exposed API endpoints enable unauthenticated attackers to automate user enumeration (/wp-json/wp/v2/users) and map backend resources for brute-force attacks.	● Low (CVSS: 3.1)

4	Permissive SPF Email Configuration	DNSRecon identified SPF record configured with soft fail (~all)	Allows unauthorized mail servers to send spoofed emails originating from @networkwalks.com with only a soft failure flag, increasing phishing delivery risks.	• Low (CVSS: 3.1)
5	DNSSEC Cryptographic Signing Inactive	WHOIS and DNSRecon confirmed DNSSEC status as 'unsigned'	Leaves DNS responses vulnerable to cache poisoning and spoofing attacks, potentially allowing malicious resolvers to redirect traffic.	• Low (CVSS: 2.6)
6	Hosting IP & Server Header Disclosures	Nslookup & cURL revealed IP 192.232.216.135 and Server: Apache	Reveals direct network location and web server software, assisting attackers in infrastructure mapping.	• Low (CVSS: 2.5)
7	Internal Subnet Host & Hypervisor Visibility	Zenmap ping sweep mapped 4 active virtual hosts with VMware OUI signatures	Internal network visibility facilitates lateral movement and reconnaissance across virtual workloads if network segmentation is absent.	• Low (Internal)

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

- **Suppress authoritative DNS daemon versions:** Configure authoritative BIND nameservers with 'version none;' inside named.conf to prevent remote version fingerprinting via CHAOS queries.
- **Sanitize web technology & CMS metadata:** Disable WordPress generator meta tags and configure Apache with 'ServerTokens Prod' and 'ServerSignature Off' to conceal web server and plugin details.
- **Restrict WordPress REST API access:** Disable or enforce authentication on sensitive REST API routes (such as /wp-json/wp/v2/users) to mitigate automated user enumeration.
- **Deploy HTTP security hardening headers:** Implement critical security headers including HSTS (Strict-Transport-Security), X-Frame-Options, X-Content-Type-Options, and Content-Security-Policy across the web server.
- **Enforce strict SPF & DMARC policies:** Transition the SPF record policy from soft fail (~all) to hard fail (-all) and publish an active DMARC record to prevent email spoofing.
- **Enable DNSSEC signing:** Configure DNSSEC keys on authoritative nameservers and submit DS records to GoDaddy to provide cryptographic origin authentication.
- **Maintain WAF monitoring & rule updates:** Keep ModSecurity enabled and tuned with updated OWASP Core Rule Sets (CRS) to actively block malicious payloads.
- **Implement internal network segmentation:** Enforce VLAN isolation and private subnet access controls on internal subnets (10.0.0.0/24) to restrict unauthorized host discovery and lateral movement.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity (W2-PM1), I used six Kali Linux tools to collect intelligence regarding networkwalks.com. The exercises demonstrated how WHOIS provides domain lifecycle and nameserver data, WhatWeb fingerprints CMS and web components, Nslookup resolves DNS mapping, Curl inspects HTTP headers and REST endpoints, Wafw00f identifies WAF defenses, and DNSRecon uncovers zone records and daemon software versions.

In the network scanning activity, I used Zenmap on Windows to discover live hosts, identify physical MAC addresses and hypervisor vendors, and generate a visual network topology on my local LAN subnet.

These exercises demonstrated that information gathering is an essential foundational stage in cybersecurity. Security professionals can uncover substantial architectural insights before attempting any active interaction. Documenting technical observations clearly with realistic risk evaluations and remediation steps is critical for effective vulnerability management.

8. Evidences Collected

The following terminal outputs and scan results were captured during the execution of Module W2-PM1 (Footprinting) and Module W2-PM5 (Zenmap Scanning):

All Evidence Screenshots are present in “evidence_screenshots” folder.

- End -